

MATRIS: TOWARD RELIABLE AND EFFICIENT PRE-TRAINED MACHINE LEARNING INTERACTION POTENTIALS

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Paper under double-blind review

ABSTRACT

Universal MLIPs (uMLIPs) demonstrate broad applicability across diverse material systems and have emerged as a powerful and transformative paradigm in chemical and computational materials science. Equivariant uMLIPs achieve state-of-the-art accuracy in a wide range of benchmarks by incorporating equivariant inductive bias. However, the reliance on tensor products and high-degree representations makes them computationally costly. This raises a fundamental question: as quantum mechanical-based datasets continue to expand, can we develop a more compact model to thoroughly exploit high-dimensional atomic interactions? In this work, we present MatRIS (Materials Representation and Interaction Simulation), an invariant uMLIP that introduces attention-based modeling of three-body interactions. MatRIS leverages a novel separable attention mechanism with linear complexity $O(N)$, enabling both scalability and expressiveness. MatRIS delivers accuracy comparable to that of leading equivariant models on a wide range of popular benchmarks (Matbench-Discovery, MatPES, Molecular dataset, DPA2 dataset, etc). Taking Matbench-Discovery as an example, MatRIS achieves the highest F1 score(0.844) while improving training efficiency by up to $13.5\times$. The work indicates that our carefully designed invariant models can match or exceed the accuracy of equivariant models at a fraction of the cost, shedding light on the development of accurate and efficient uMLIPs.

1 INTRODUCTION

Quantum Mechanism (QM)-based calculations are the cornerstone of modern drug and material research, providing highly accurate modeling of interatomic interactions. However, its prohibitive computational cost makes large-scale simulations intractable (De Vivo et al., 2016; Jain et al., 2013b; Merchant et al., 2023). Machine learning interatomic potentials (MLIPs) have emerged as a powerful alternative, enabling accelerated, long-timescale molecular dynamics (MD) simulations while retaining near-quantum-chemical accuracy. With the increase in QM-based reference data and model innovations, MLIPs have demonstrated remarkable accuracy and generalization in property prediction and materials discovery (Merchant et al., 2023; Barroso-Luque et al., 2024; Zhang et al., 2024; Yang et al., 2024; Zhang et al., 2025; Fu et al., 2025; Wood et al., 2025).

Graph neural networks (GNNs) have been widely adopted for 3D molecular modeling,

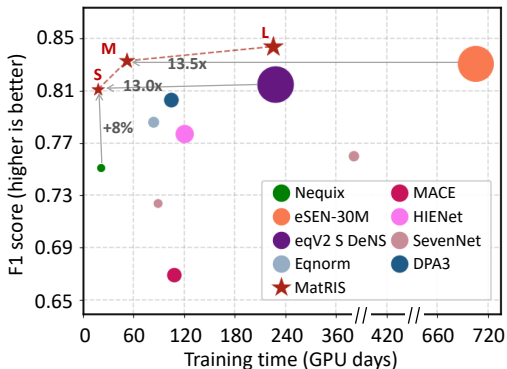


Figure 1: Trade-offs between training time and F1 score of uMLIPs. Training times for eSEN-30M and eqV2 S DeNS are estimated on an Nvidia A100 GPU. Nequix(Koker & Smidt, 2025) was trained on JAX; all others on PyTorch. Larger marks indicate models with more parameters.

where atoms are represented as nodes and interatomic interactions as edges (Qu & Krishnapriyan, 2024; Liao & Smidt, 2023; Liao et al., 2024b). Through Message Passing (MP), node features are iteratively updated to capture local and global structural interactions. To enhance model expressiveness and generalization, many MLIPs incorporate domain-specific inductive biases (e.g., translation, rotation, permutation, reflection invariance, or equivariance). Depending on how these symmetries are encoded, MLIPs are broadly divided into invariant and equivariant architectures (Duval et al., 2024; Jacobs et al., 2025). In invariant models, the structural descriptor is encoded based on attributes such as interatomic distances, bond angles, and dihedral angles (Gasteiger et al., 2022a;b; 2024; Deng et al., 2023; Zhang et al., 2025). In equivariant models, the predominant strategy for enforcing higher-order equivariance relies on computationally intensive tensor products of rotation order L (Batzner et al., 2022; Batatia et al., 2023; Liao & Smidt, 2023; Liao et al., 2024b).

Prior work demonstrates that equivariant models often deliver superior accuracy (Batzner et al., 2022). Meanwhile, the results in a popular benchmark (Matbench-Discovery leaderboard (Riebesell et al., 2025)) indicate that equivariant GNNs achieve higher accuracy. When the training data is MPTrj, eSEN-30M-MP (Fu et al., 2025) and eqV2 S DeNS (Barroso-Luque et al., 2024) reach F1 scores of 0.831 and 0.815, with energy errors of 0.033 and 0.036 meV/atom, respectively. Despite the accuracy gains, the heavy tensor product operations make equivariant methods significantly more computationally expensive and memory-intensive. Our investigation into the training cost of several mainstream pretrained models is summarized in Fig.1. eSEN-30M-MP and eqV2 S DeNS show superior F1 score while requiring 705 and 228 GPU days, respectively. This high computational demand can be caused by three main factors: 1) the intensive equivariant operations such as tensor products, 2) the large number of model parameters, and 3) the prolonged training schedules (e.g., 100, 150, and 600 training epochs are reported for eSEN-30M-MP, eqv2 S DeNS, and SevenNet-13i5 (Kim et al., 2024), respectively).

Incorporating equivariance into the GNN MLIPs serves as an implicit form of data augmentation. AlphaFold has shown that, with sufficient data, non-equivariant models can accurately predict protein secondary structures (Jumper et al., 2021; Abramson et al., 2024). In MLIPs, the rapid increase in QM-based reference datasets (Barroso-Luque et al., 2024; Levine et al., 2025) motivates us to ask: **Is the equivariance indispensable as the QM-based dataset continues to increase? Can we develop a more compact architecture to capture the high-dimensional atomic interactions encoded in QM-based data sufficiently?**

We have the following findings: 1) on the invariant model. Recent studies (Neumann et al., 2024; Qu & Krishnapriyan, 2024; Rhodes et al., 2025) show that invariant models allow more flexibility in exploring the potential energy surface while being computationally more efficient. 2) on a more compact architecture (an architecture that can fully exploit QM-based data). Element types and pairwise interactions have been shown to be insufficient for distinguishing graphs with different chemical properties (Xu et al., 2019). Incorporating three-body interactions is needed to exploit the knowledge in QM-based data. Self-attention mechanisms (Xu et al., 2019; Liao & Smidt, 2023; Liao et al., 2024b; Qu & Krishnapriyan, 2024) have proven to be a promising method in improving model expressiveness, also benefiting in model scalability.

Building on these insights, we introduce an invariant MLIP: interatomic potential for **Materials Representation and Interaction Simulation (MatRIS)**. To the best of our knowledge, our model is the first to leverage attention mechanisms for modelling three-body interaction. MatRIS consists of graph generation, feature embedding, graph attention, refinement, and a readout block. We provide the ablation study of these modules in this paper. Putting all these modules together, MatRIS achieves SOTA accuracy and efficiency across a wide range of chemical applications. Additionally, the novel separable attention has lower complexity ($O(N)$) compared to full attention ($O(N^2)$). Across diverse benchmarks, MatRIS achieves competitive results. MatRIS-L achieves state-of-the-art (SOTA) results on compliant Matbench-Discovery with an F1 of **0.844** and a root mean square displacement (RMSD) of **0.727**. Moreover, MatRIS-S and MatRIS-M deliver accuracy comparable to eqV2 S DeNS and eSEN-30M-MP, respectively, while improving training efficiency by **13.0×** and **13.5×**, respectively. These results demonstrate MatRIS’s strong potential for applications in materials science and drug discovery.

2 BACKGROUND

Invariant MLIPs. Invariants can be met by employing internal coordinates instead of Cartesian coordinates, because internal coordinates, such as interatomic distances, angles, and dihedral angles, are invariant with respect to rotations and translations of the system Behler (2014a). Early invariant MLIPs employ handcrafted invariant polynomials to fulfill the invariant criteria Behler (2011; 2014b; 2015); Smith et al. (2017); Novikov et al. (2020); Fan et al. (2022). DeePMD encodes the feature matrix D by $G^T R R^T G$, where G is learned by a local embedding network and R is the relative coordinates matrix Zhang et al. (2018). Recently, GNN-based invariant MLIPs have emerged. DimeNet Gasteiger et al. (2020) introduced directional MP, allowing angular information to be embedded in edge updates between atoms. GemNet Gasteiger et al. (2024) incorporated dihedral angles, and SphereNet Liu et al. (2022) further integrated torsional information. ComENet Wang et al. (2022) introduces a completeness-aware MP scheme. Invariant MLIPs are gradually increasing representational expressiveness while maintaining the inherent efficiency.

Equivariant MLIPs. Equivariance is a powerful inductive bias that enhances data efficiency and generalization. Equivariant MLIPs can be broadly divided into scalarization-based methods Schütt et al. (2021); Du et al. (2023); Thölke & Fabritius (2022) and high-degree steerable methods Zhou et al. (2024); Batzner et al. (2022); Batatia et al. (2023); Liao & Smidt (2023). Here, we focus on high-degree steerable equivariant MLIPs, which encode features using irreducible representations (irreps), ensuring equivariance to 3D rotations. Each irrep of degree L corresponds to a $2L + 1$ -dimensional vector space. In equivariant GNN-based MLIPs, MP involves transforming and combining these type- L vectors. To interact across degrees during MP, tensor products (by using Clebsch–Gordan coefficients to combine) are employed. These models are limited to small degrees of equivariant representations due to their computational complexity. Recent studies, Passaro & Zitnick (2023); Liao et al. (2024b); Aykent & Xia (2025) have focused on improving the efficiency of equivariant operations by replacing or approximating tensor products. Equivariant MLIPs continue to deliver SOTA accuracy on various benchmarks, while remaining computationally demanding.

3 MATRIS

In this section, we introduce the detailed architecture of MatRIS (see Fig. 3(a)). The interaction between the atom graph and the line graph is described in Sec.3.1. The Graph Attention module is depicted in Sec.3.2. In Sec.3.3, we describe other key components of MatRIS. An overview of MatRIS is shown in Fig.3, and the model formalizations are detailed in Appendix.A.

3.1 LINE–ATOM GRAPH INTERACTION

To model three-body interactions, we explicitly construct a Line Graph. In the Atom Graph, nodes represent atom types and edges represent pairwise interactions (bonds), whereas in the Line Graph, nodes represent edges of the Atom Graph and edges represent three-body interactions (angles) (Harary & Norman, 1960; Whitney, 1992).

Specifically, given an atom graph $G_A = (V_A, E_A)$, where V_A is the set of atoms and E_A is the set of edges within a cutoff distance r_{cut}^L , the corresponding Line Graph $G_L = (V_L, E_L)$ is constructed as follows: ① Each node in V_L corresponds to an edge in E_A ; ② An edge $e \in E_L$ is added between two nodes if their corresponding edges in E_A share a common atom, representing the angular information formed by the three atoms (i.e., the three-body interaction). The conversion from the Atom Graph to the Line Graph is illustrated in Fig. 2.

For graph information fusion, we first update the Line Graph to obtain edge and angular features encoding three-body interactions. The updated edge features are then propagated back to the Atom Graph, allowing atomic features to incorporate higher-order information from the Line Graph.

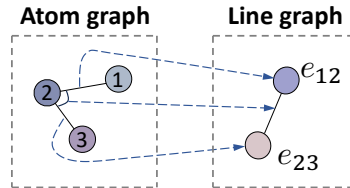


Figure 2: Conversion from an atom graph to a line graph.

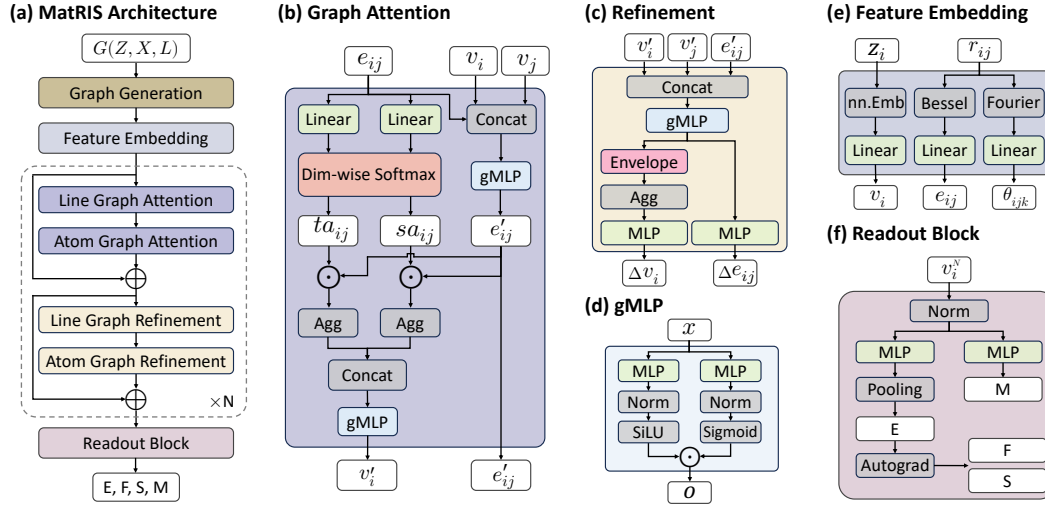


Figure 3: Overview of MatRIS. The model architecture (a) consists of feature embedding (e), graph attention (b), refinement (c), and a readout block(f).

3.2 GRAPH ATTENTION

Dim-wise Softmax. Recent studies have shown that attention mechanisms play an important role in improving both the accuracy and scalability of MLIPs (Qu & Krishnapriyan, 2024). Existing approaches (Zhang et al., 2024; Liao & Smidt, 2023; Qu & Krishnapriyan, 2024) typically compute attention scores α_{ij} to weight the value vectors $V \in \mathbb{R}^{\text{neighbors} \times d}$. The same weights α_{ij} are applied to all feature dimensions d , implicitly assuming equal importance across dimensions. However, this assumption limits the model’s ability to distinguish the independent contributions of different feature channels. Our proposed Dim-wise Softmax computes attention scores independently for each feature dimension. Given an input feature $x \in \mathbb{R}^{\text{neighbors} \times d}$ and a neighbor list $\mathcal{N}$, the Dim-wise Softmax is computed as follows:

$$\alpha_{ij} = \text{Dim-wise Softmax}(x_{ij}, \mathcal{N}(i)) = \frac{\exp(x_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(x_{kj})} \quad (1)$$

where $\alpha \in \mathbb{R}^{\text{neighbors} \times d}$ is the attention weight matrix, x_{ij} denotes the j -th feature of node i , and $\mathcal{N}(i)$ represents the set of neighboring nodes of node i . This approach preserves the independence of feature dimensions while emphasizing the relative importance of different neighbors in each dimension, thereby enhancing the model’s ability to capture local structural information.

Separable Attention. In molecular systems, interatomic interactions are directional, and each atom plays two roles: target node and source node. Most existing methods only aggregate information from the source node to the target node (Gasteiger et al., 2022b; Liao & Smidt, 2023; Liao et al., 2024b), which assumes symmetric information flow. However, this is not always true in real physical systems. For example, in polar bonds, charged environments, or local defect structures, the effect of the source node on the target node can differ from the effect of the target node on the source node (Bengtsson, 1999; Kühne & Khaliullin, 2013). To address this, we introduce two independent sets of attention weights: source attention weights and target attention weights. The first models how neighboring atoms affect the central atom, while the second captures how the central atom influences its neighbors. In this way, the two roles of atoms are explicitly separated during aggregation. The overall workflow is illustrated in Fig. 3(b). Given the interaction e_{ij} between a target node v_i and a source node v_j , we compute the attention weights as follows:

$$t_{ij} = \text{Linear}(e_{ij}) \quad \text{and} \quad ta_{ij} = \text{Dim-wise Softmax}(t_{ij}, \mathcal{N}(i)) \quad (2)$$

$$s_{ij} = \text{Linear}(e_{ij}) \quad \text{and} \quad sa_{ij} = \text{Dim-wise Softmax}(s_{ij}, \mathcal{N}(j)) \quad (3)$$

Here, $N(i)$ and $N(j)$ denote the indices of the target and source nodes, respectively. The final attention outputs are obtained as the weighted sum of ta_{ij} , sa_{ij} , and e'_{ij} . Here, e'_{ij} is obtained by concatenating e_{ij} , v_i , and v_j , followed by target and source feature fusion through a gMLP (see Fig. 3(d)). The two attention branches share the same computational flow and can therefore be executed in parallel. We also implement optimized kernels to improve training efficiency.

3.3 OVERALL ARCHITECTURE

Graph Generation. Giving a crystalline system $G(Z, X, L)$, where $Z \in \mathbb{R}^n$ denotes the atomic numbers, $X \in \mathbb{R}^{n \times 3}$ represents the atomic coordinates, and $L \in \mathbb{R}^{3 \times 3}$ refers to the lattice. We first perform periodic repetition of the structure and then employ a radius-cutoff graph construction to represent it. Inspired by CHGNet(Deng et al., 2023), we construct the atom graph G^a (with atoms as nodes and bonds as edges) based on r_a and the line graph G^b (with bonds as nodes and angles as edges) based on r_b for the crystal structure.

Feature Embedding. Atomic numbers are initialized using trainable embeddings, while pairwise distances are encoded via a learnable radial Bessel basis with envelope functions (Gasteiger et al., 2022a). For bond angles, we employ a learnable Fourier basis expansion following (Deng et al., 2023). These input features are then passed through a linear layer to generate initial node, edge, and three-body representations. This block is illustrated in Fig. 3(e).

Refinement. The output of Graph Attention v'_i and e'_{ij} is concatenated and processed by a gMLP operator (shown in Fig. 3(c)). This representation is further refined by an Envelope function (Gasteiger et al., 2022a) (ensures smoothness). The refined signals are then aggregated over neighboring interactions, enabling each edge to integrate local information. Finally, the output is transformed by a MLP operator to produce the residual node features Δv_i and edge features Δe_{ij} .

Readout Block. The Readout Block (Fig. 3(f)) takes the node features v_i from the last MP layer. After the LayerNorm operation, the features are passed through MLPs to predict node energies and magnetic moments (M). Atomic energies are summed to obtain total energy (E). To ensure the reliability (Bigi et al., 2025), atomic forces (F_i) and stress (σ) are computed by Eq. 4. V is volume.

$$F_i = -\frac{\partial E}{\partial X_i}, \sigma = \frac{1}{V} \frac{\partial E}{\partial \epsilon} \quad (4)$$

4 EXPERIMENTS

We evaluated the MatRIS on Matbench-Discovery benchmark (Sec.4.1), MatPES benchmark (Sec.4.2), Molecular zero-shot benchmark (Sec.4.3), DPA2 test sets (Sec.4.4). The best results are in bold, and the second best are underlined. The training strategies are introduced in Appendix B, and the training hyperparameter settings are detailed in Appendix C. Experiments on Matbench-Discovery results for full and 10k most stable splits are referred to in Appendix D.2. Moreover, we conducted ablation studies on the network modules and training methods (Sec.4.5). Finally, we analyzed the efficiency of MatRIS (Sec.4.6). In addition, the experiments on the Zeolite benchmark and DOS results are provided in Appendix D.

4.1 MATBENCH DISCOVERY

Dataset and Setting. The Matbench-Discovery benchmark (Riebesell et al., 2025) is a well-established benchmark for evaluating the ability of models in new material discovery. In this benchmark, all models are required to optimize the geometry and predict the formation energy of each of the 256k structures in the WBM test set (Wang et al., 2021). These results are then used to assess thermodynamic stability at the ground state (0 K). Following the ‘‘compliant’’ setting, models are trained on the MPTrj dataset (Deng et al., 2023), which contains 1.6 million configurations. We use three variants of MatRIS—S, M, and L. The models predict total energy, forces, stress, and magnetic moments of the system. More details on hyper-parameters are given in Table.7 in appendix. Moreover, inspired by (Zaidi et al., 2022; Liao et al., 2024a), we apply denoising pretraining to MatRIS-M and MatRIS-L. The specific details are described in Appendix C. Notably, this denoising

Table 1: MatRIS performance on the compliant Matbench-Discovery benchmark with results on unique structure prototypes. ‘↑’/‘↓’ stands for higher/lower is better. All models accessed before September 24, 2025.

Model	Param.	F1↑	DAF↑	Precision↑	Recall↑	Accuracy↑	MAE↓	R2↑	K_{srme} ↓	RMSD↓
CHGNet	0.41M	0.613	3.361	0.514	0.758	0.851	0.063	0.689	1.717	0.0949
MACE-MP-0	4.69M	0.669	3.777	0.577	0.796	0.878	0.057	0.697	0.647	0.0915
GRACE-2L	15.3M	0.691	4.163	0.636	0.757	0.896	0.052	0.741	0.525	0.0897
Allegro-MP-L	18.7M	0.751	4.516	0.690	0.823	0.915	0.044	0.778	0.504	0.0816
Nequix MP	0.71M	0.751	4.455	0.681	0.836	0.914	0.044	0.782	0.446	0.0853
SevenNet-l3i5	1.17M	0.760	4.629	0.708	0.821	0.920	0.044	0.776	0.55	0.0847
NequIP-MP-L	9.6M	0.761	4.704	0.719	0.809	0.921	0.043	0.791	0.452	0.0856
ORB v2 MPtrj	25.2M	0.765	4.702	0.719	0.817	0.922	0.045	0.756	1.725	0.1007
HIENet	7.51M	0.777	4.932	0.754	0.801	0.929	0.041	0.793	0.642	0.0795
Eqnorm MPtrj	1.31M	0.786	4.844	0.741	0.838	0.929	0.040	0.799	0.408	0.0837
DPA-3.1-MPtrj	4.81M	0.803	5.024	0.768	0.841	0.936	0.037	0.812	0.650	0.0801
eqV2 S DeNS	31.2M	0.815	5.042	0.771	0.864	0.941	0.036	0.788	1.676	0.0757
eSEN-30M-MP	30.1M	0.831	5.260	0.804	<u>0.861</u>	0.946	<u>0.033</u>	<u>0.822</u>	0.340	0.0752
MatRIS-S	4.3M	0.811	5.127	0.784	0.840	0.940	0.036	0.803	0.730	0.0766
MatRIS-M	6.3M	<u>0.833</u>	<u>5.363</u>	<u>0.820</u>	0.847	<u>0.948</u>	<u>0.033</u>	<u>0.820</u>	0.542	<u>0.742</u>
MatRIS-L	10.4M	0.844	5.431	0.830	0.858	0.951	0.031	0.829	0.505	0.727

is also adopted in several other works (ORB v2 MPtrj (Neumann et al., 2024), eqV2 S DeNS, and eSEN-30M-MP). Structures are relaxed using MatRIS and the FIRE (Bitzek et al., 2006) optimizer, with convergence reached after 500 steps or when the maximum force falls below 0.05 eV/Å.

Results and Analysis. We summarize the comparison with other models in Table.1 (as of September 24, 2025). MatRIS-S/M achieves performance comparable to eqv2 S DeNS/eSEN-30M-MP while using fewer parameters and lower computational costs (See Fig.1 for a comparison of training cost). These results demonstrate the effectiveness of MatRIS. Moreover, MatRIS-L achieves state-of-the-art (SOTA) performance across all metrics, with an F1 score of 0.844. It also attains a RMSD of 0.727 when comparing relaxed structures to DFT reference values. We also report the results of MatRIS on full testset and 10k most stable (see Table.9 and Table.10 in Appendix D.2).

4.2 MATCALC BENCHMARK

Table 2: MatRIS-MatPES performance on the MatCalc benchmark.

Model	Param.	Equilibrium		Near-equilibrium				
		d ↓	Ef ↓	K ↓	G ↓	CV ↓	f/f_{DFT} ↑	
M3GNet	0.66M	<u>0.42</u>	0.11	26	25	27	0.97	
CHGNet	2.7M	0.43	0.082	24	21	23	0.91	
TensorNet	0.84M	0.37	<u>0.081</u>	<u>18</u>	<u>15</u>	13	0.93	
MatRIS	1.4M	0.54	0.068	15	13	<u>16</u>	<u>0.96</u>	

Table 3: MatRIS-MPTrj performance on the MatCalc benchmark.

Model	Param.	Equilibrium		Near-equilibrium				
		d ↓	Ef ↓	K ↓	G ↓	CV ↓	f/f_{DFT} ↑	
CHGNet	0.43M	0.51	0.092	17.0	30.0	24.0	0.830	
MACE-L	5.7M	0.43	–	25.0	<u>22.5</u>	11.6	0.829	
SevenNet-l3i5	1.17M	0.55	0.057	<u>13.2</u>	170.2	8.03	0.922	
eqV2 S DeNS	31.2M	0.25	0.033	27.2	29.1	25.9	0.964	
eSEN-30M-MP	30.1M	0.34	<u>0.039</u>	18.8	77.3	4.66	0.986	
MatRIS-M	6.3M	<u>0.32</u>	0.041	12.4	16.4	<u>7.39</u>	<u>0.983</u>	

Dataset and Settings. MatCalc benchmark (Kaplan et al., 2025) covers equilibrium properties (relaxed structure similarity d , formation energy E_f), near-equilibrium properties (bulk modulus K , shear modulus G , constant-volume heat capacity C_V , force softening f/f_{DFT}), and is constructed from test data collected from the Materials Project (Jain et al., 2013a), Alexandria (Schmidt et al., 2024), WBM high-energy states (Wang et al., 2021). We compare models trained on MatPES-PBE (Kaplan et al., 2025) and MPTrj (Deng et al., 2023), with the results reported in Table. 2 and Table. 3, respectively. The training parameters are listed in Appendix Table 7.

Results and Analysis. Table 2 summarizes the performance of models trained on MatPES-PBE, with MatRIS achieving the overall best results and clearly outperforming others in the prediction of formation energy E_f . For models trained on MPTrj (Table 3), MatRIS achieves SOTA or near-SOTA performance on 83% of the metrics. Notably, MatRIS remains robust on “Near-equilibrium” tasks regardless of the training dataset. Although the higher fraction of near-equilibrium structures in MPTrj amplifies PES “Softening” (f/f_{DFT}) (Deng et al., 2024), MatRIS maintains stable performance. Comparison with CHGNet further shows that changing the dataset does not lead to significant softening, highlighting the effectiveness of the architecture design.

4.3 MOLECULAR ZERO-SHOT BENCHMARK

Table 4: MatRIS performance on the Molecular zero-shot benchmark. Energy (E) is reported in kcal/mol and Force (F) in kcal/mol/Å.

Model	TorsionNet-500				MD22 (Ac-Ala3-NHMe)		ANI-1x		AIMD-Chig	
	E MAE	E RMSE	MAEB ^a	NABH _h ^b	E	F	E	F	E	F
AIMNet2	0.38	0.55	0.58	82	-	-	-	-	-	-
DPA2-Drug	0.24	0.35	0.36	18	-	-	-	-	-	-
MACE-OFF-L	0.14	0.21	0.23	4	0.041	5.205	0.406	3.948	0.050	3.853
DPA3-L24	<u>0.06</u>	<u>0.09</u>	<u>0.09</u>	0	<u>0.040</u>	<u>5.154</u>	0.873	37.45	<u>0.038</u>	<u>3.749</u>
MatRIS-M	0.04	0.07	0.07	0	0.031	5.140	0.301	<u>4.414</u>	0.009	3.604

^a The MAE of the torsional barrier height is defined as the energy difference between the minimum and maximum along the torsional rotation.

^b The number of molecules with barrier height errors exceeding 1 kcal/mol.

Dataset and Settings. AIMNet and DPA2-Drug were trained on the datasets from Ref. (Anstine D, 2023) and Ref. (Yang et al., 2025), respectively, while the other models were pre-trained on SPICE-MACE-OFF23 (Kovács et al., 2025), which contains 951,005 small-molecule configurations. The energies and forces of all configurations were computed at the ω B97M-D3(BJ)/def2-TZVPPD level. Following (Peng et al., 2025), we selected the Ac-Ala3-NHMe from the MD22(Chmiela et al., 2023), ANI-1x(Smith et al., 2018), and AIMD-Chig(Wang et al., 2023) datasets, which contain 2,212, 8,861, and 19,800 configurations, respectively. The training parameters can be found at Table 7 in appendix. For the MD22, ANI-1x, and AIMD-Chig tests, we have eliminated errors arising from inconsistencies between DFT functionals.

Results and Analysis. We selected available molecular pre-trained models (MACE-OFF-L (Kovács et al., 2025) and DPA3-L24 (Zhang et al., 2025)) for comparison, and the results are summarized in Table. 4. MatRIS-M performs well across four downstream datasets. On the TorsionNet-500 test set, its energy prediction error is reduced by about 22.2%–33.3% compared to the current SOTA model DPA3. In the remaining three datasets, MatRIS-M ranks first in five out of six metrics and second in the remaining one, demonstrating that it effectively leverages the knowledge from the pre-training dataset to achieve more reliable performance on downstream tasks.

4.4 DPA2 TEST SET

Dataset and Settings. We evaluate the performance of the MatRIS model on the DPA2 dataset (Zhang et al., 2024). This composite dataset integrates 18 domain-specific subsets (e.g., Alloy, Drug, H₂O, OC2M) and is generated using various DFT software (e.g., VASP, Gaussian,

Table 5: MatRIS performance on DPA2 test sets from (Zhang et al., 2024). Energy(**E**) RMSE is in **meV/atom**, force(**F**) RMSE is in **meV/Å**. ‘OOM’ indicates an Out-Of-Memory errors. ‘NC’ indicates that the model employs a non-conservative force prediction.

	GemNet-OC ^{NC}		NequIP		MACE		eqV2 ^{NC}		DPA3-L24		MatRIS-M	
Dataset	E	F	E	F	E	F	E	F	E	F	E	F
Alloy	14.3	<u>85.1</u>	44.0	175.6	16.2	190.2	8.5	62.7	<u>7.1</u>	99.2	5.8	96.2
Cathode-P	1.5	17.9	14.3	69.8	2.6	37.8	1.1	<u>14.9</u>	<u>0.6</u>	18.6	0.3	11.9
Cluster-P	47.7	69.6	75.1	216.6	41.3	189.7	34.6	<u>104.4</u>	<u>29.3</u>	118.3	12.8	113.9
Drug	40.5	93.6	21.6	187.2	-	-	29.8	807.4	<u>5.5</u>	<u>54.2</u>	1.4	39.7
FerroEle-P	1.5	17.9	1.1	23.0	2.3	31.7	1.1	<u>13.0</u>	<u>0.3</u>	13.4	0.2	10.9
OC2M	25.0	129.1	97.4	226.1	-	-	<u>6.7</u>	45.2	9.0	128.0	5.9	<u>77.9</u>
SSE-PBE-P	2.7	8.2	1.6	41.1	1.8	29.9	OOM	OOM	<u>0.5</u>	<u>19.8</u>	0.3	11.4
SemiCond	8.0	<u>94.4</u>	20.5	180.7	12.7	182.8	3.9	40.8	<u>3.4</u>	108.3	2.7	95.0
H2O-PD	OOM	OOM	0.9	27.1	79.9	29.7	OOM	OOM	<u>0.4</u>	<u>13.7</u>	0.3	10.7
AgUAu-PBE	106.0	<u>8.0</u>	42.3	43.8	369.1	34.5	23.4	4.4	1.1	10.9	<u>2.5</u>	10.7
AlUMgUCu	5.9	<u>9.4</u>	38.0	48.3	7.7	42.9	<u>1.9</u>	5.7	2.0	13.0	1.2	15.9
Cu	6.1	<u>5.8</u>	6.2	16.7	38.8	13.6	<u>1.7</u>	3.8	<u>1.7</u>	7.2	0.5	6.1
Sn	8.4	<u>33.7</u>	18.2	62.2	-	-	5.2	19.6	<u>2.9</u>	49.8	2.3	45.9
Ti	44.5	87.9	27.6	137.4	8.3	94.2	19.1	48.6	<u>4.1</u>	91.7	3.3	<u>80.2</u>
V	17.9	79.3	8.8	91.6	14.2	140.4	5.6	47.4	<u>2.8</u>	71.8	1.9	<u>62.9</u>
W	79.1	81.2	20.8	160.4	15.6	181.2	46.8	51.3	<u>2.5</u>	83.3	1.9	<u>66.3</u>
C12H26	125.8	<u>518.7</u>	121.4	715.6	81.9	802.3	123.1	907.4	<u>42.4</u>	541.6	19.5	287.6
HfO2	1.2	16.1	1.5	58.8	2.3	14.7	<u>1.0</u>	9.1	1.4	28.9	0.3	<u>14.6</u>

ABACUS). The training set consists of 4,045,094 structures, while the test set contains 1,074,285 structures. We evaluate MatRIS-M using the same training configurations and maintain consistency with the training and test sets as in (Zhang et al., 2024) to ensure comparability. More detailed hyper-parameters are reported in Table.7 in Appendix C.

Results and Analysis. All results are summarized in Table.5, with benchmark data for other models taken from (Zhang et al., 2024; 2025). Overall, MatRIS-M achieves better accuracy in both energy and force predictions, demonstrating its robustness across diverse domains. Notably, however, eqV2 and GemNet-OC exhibit lower force errors than MatRIS in certain systems. This advantage likely stems from their non-conservative force prediction, where energy and forces are fitted separately at the expense of physical consistency.

4.5 ABLATION STUDY

We conduct ablation studies on various modules of MatRIS (see 4.5) and their corresponding training methods (see 4.5). We train the MatRIS-S model on MPTrj (Deng et al., 2023) and evaluate it on 15,000 randomly sampled structures from the WBM (Wang et al., 2021) dataset, reporting performance using the MAE of formation energies.

Module Ablation. We evaluated the impact of Dimwise Softmax, Separable Attention, and Learnable Envelope on model accuracy, as shown in Table.6(a). Specifically, replacing Dimwise Softmax with standard softmax (i.e., sharing the same weight across all feature dimensions) increased the MAE to 28.4 meV/atom. Subsequently, performing attention-based aggregation only in the source-to-target direction further raised the MAE. Finally, replacing the learnable envelope function with a fixed one also degraded performance.

Training Method Ablation. We further analyze the contributions of different training strategies to model performance, as shown in Table. 6(b). Pretraining with a denoising (see Sec. B.3 in Appendix) effectively improves performance, as predicting noise helps mitigate over-smoothing (Godwin et al., 2022; Zaidi et al., 2022). Notably, predicting magmoms also enhances accuracy, since magmoms

Table 6: Ablation studies. Formation energy (Ef) MAE is in meV/atom; lower is better.

Dimwise softmax	Separable attention	Learnable envelop	Ef (MAE)	denoising pretraining	w/o Magmom	Graph-level loss	Ef (MAE)
✓	✓	✓	28.0	✓	✓	✓	27.2
✗	✓	✓	28.4	✗	✓	✓	28.0
✗	✗	✓	29.1	✗	✗	✓	29.7
✗	✗	✗	31.3	✗	✗	✗	30.2

(a) Effect of modules: dimwise-attention, separable attention and learnable envelop.

(b) Effect of training methods: denoising pretraining, magnetic moment prediction, and graph-level loss.

help distinguish features in different chemical environments (Deng et al., 2023) and, as a node-level task, likely reduce over-smoothing during training. Finally, performing loss reduction at the graph level prevents the force loss from being biased by differences in system size (see Sec. B.1 in Appendix for more detail), further improving performance.

4.6 EFFICIENCY-ACCURACY ANALYSIS.

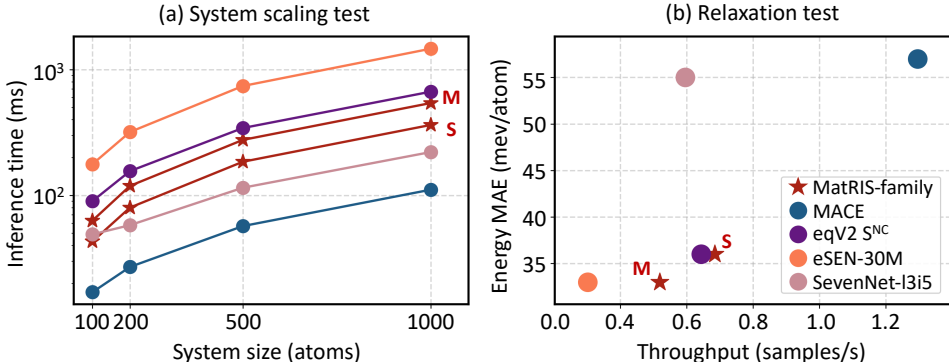


Figure 4: Efficiency-accuracy comparison.

In this section, we compare the computational efficiency of MatRIS with several other mainstream models. First, the inference time on different systems (excluding graph construction) is summarized in Fig. 4(a). MatRIS shows higher efficiency than eqV2 and eSEN, but lower than MACE-L, because MACE-L has only 2 layers while MatRIS-S and MatRIS-M have 4 and 6 layers, respectively. We also evaluate the models in practical applications. We randomly select 500 structures from the WBM dataset (Wang et al., 2021) and perform relaxations using ASE, recording the throughput and energy MAE for each model. As shown in Fig. 4(b), MatRIS achieves a good balance between speed and accuracy, maintaining high precision while offering faster computation.

5 CONCLUSION AND FUTURE WORK

In this work, we review the characteristics of invariant and equivariant MLIPs in the era of rapidly expanding QM-based datasets. Equivariant uMLIPs deliver superior accuracy while their reliance on tensor products and high-degree representations leads to prohibitive computational costs. Motivated by the question of whether strict equivariance remains indispensable, we introduce MatRIS, an invariant uMLIP that leverages attention to model three-body interactions. Across multiple benchmarks, MatRIS attains SOTA results, opening a new path toward accurate and efficient MLIPs.

In future work, we will scale MatRIS to even larger QM-based datasets to further validate its expressiveness. We aim to develop a reliable distillation strategy to develop student MLIPs. We also plan to expand MatRIS to incorporate long-range electrostatics, thereby enhancing its applicability to more complex downstream tasks.

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APPENDIX

A Details of MatRIS

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A DETAILS OF MATRIS

For completeness, in this section we present the details of MatRIS. Given a material graph $G = (Z, X, L)$, the atomic number list is denoted as $Z \in \mathbb{R}^{n \times 1}$, the atomic positions as $X \in \mathbb{R}^{n \times 3}$, and the lattice as $L \in \mathbb{R}^{3 \times 3}$. In **Graph Generation** (see Fig.5), we first perform periodic repetition, and then construct the Atom Graph G_A and the Line Graph G_L based on radial cutoffs r_a and r_l , respectively. We typically set $r_l < r_a$ for computational efficiency.

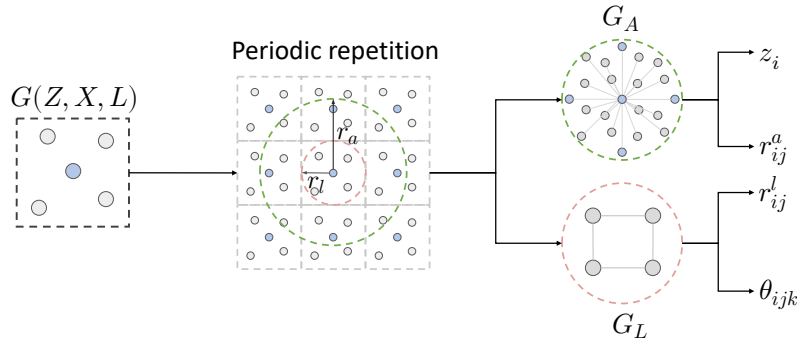


Figure 5: The detail of Graph Generation.

In **Feature Embedding**, the atomic types z_i are expanded using a learnable embedding matrix $A \in \mathbb{R}^{\text{max_element} \times d}$ (see Eq. 5). Following (Gasteiger et al., 2022a), the pairwise distances r_{ij}^a of G_A are expanded with Bessel basis functions, and smoothed with an envelope function ($\mu(r_{ij})$) to ensure that pairwise interactions decay smoothly to zero at the cutoff radius, as shown in Eq. 6. For the initialization of three-body interactions θ_{ijk} , we follow (Deng et al., 2023) and employ the computationally simpler Fourier basis instead of spherical harmonics (we also tested spherical harmonics but observed no change in accuracy), as shown in Eq. 7. Finally, all initialized features undergo nonlinear transformations and are projected into a higher-dimensional space.

$$v_i^0 = W(z_i A) + b \quad (5)$$

$$e_{ij}^0 = \mu(r_{ij}^a) \cdot \sqrt{\frac{2}{r_a}} \frac{\sin(\frac{n\pi}{r_a} r_{ij}^a)}{r_{ij}^a} \quad (6)$$

$$d_{ijk,m}^0 = \begin{cases} \frac{1}{\sqrt{\pi}} \cos[m\theta_{ijk}], & m \in [0, M_{max}/2], \\ \frac{1}{\sqrt{\pi}} \sin[(m - M_{max}/2)\theta_{ijk}], & m \in [M_{max}/2 + 1, M_{max}]. \end{cases} \quad (7)$$

In the **Graph Attention** module, we consider the n -th layer, which contains the Atom Graph G^A with features v_i^n and e_{ij}^n , and the Line Graph G^L with features e_{ij}^n and θ_{ijk}^n . The two graphs share the edge features e_{ij}^n . To fuse information between the graphs, we first update G^L and then incorporate its output into G^A to aid atomic feature updates (see Eq. 8). Specifically, e_{ij}^n and θ_{ijk}^n are input to the ‘Line Graph Attention’ module, producing $e'_{ij,l}(n)$ and $\theta'_{ijk}(n)$. The updated edge feature $e'_{ij,l}(n)$ encodes three-body interactions and is combined with v_i^n as input to the ‘Atom Graph Attention’ module, yielding $v'_i(n)$ and $e'_{ij}(n)$.

$$\begin{aligned} e'_{ij,l}(n), \theta'_{ijk}(n) &\leftarrow \text{Line Graph Attention}(e_{ij}^n, \theta_{ijk}^n) \\ v'_i(n), e'_{ij}(n) &\leftarrow \text{Atom Graph Attention}(v_i^n, e'_{ij,l}(n)) \end{aligned} \quad (8)$$

Notably, in G^A , atoms are treated as nodes and pairwise distances as edges, whereas in G^L , pairwise distances are treated as nodes and three-body interactions as edges. Therefore, the computation of the Line Graph Attention and Atom Graph Attention follows the same computation, differing only in their inputs. Here, we take the update of G^A as an example and present the detailed operations of the Graph Attention module.

Given the input of the current attention layer, v_i and e_{ij} (for simplicity, we denote v_i^n and $e'_{ij,l}(n)$ as v_i and e_{ij}), we first generate the fusion feature:

$$e'_{ij} = \text{gMLP}(v_i || v_j || e_{ij}) \quad (9)$$

Meanwhile, e_{ij} undergoes two nonlinear transformations. It is then passed through Dim-wise Soft-max to get the attention weights for the source and target nodes, sa_{ij} and ta_{ij} . These weights are then element-wise multiplied with e'_{ij} and fused to produce the output of the attention layer, v'_i , as shown in the equation:

$$\begin{aligned} tv_i &= \sum_{k \in \mathcal{N}(i)} (ta_{kj} \odot e'_{kj}), \quad sv_i = \sum_{k \in \mathcal{N}(j)} (sa_{kj} \odot e'_{kj}) \\ v'_i &= \text{gMLP}(tv_i || sv_i) \end{aligned} \quad (10)$$

After the Attention layer, we employ a **Refinement** layer to further enhance the geometric encodings. In this module, we also apply a learnable envelope function to smooth the potential energy surface. As before, we first update G_L and then G_A . Taking the update of G_A as an example, the inputs are v'_i and e'_{ij} , and we fuse the features from the attention layer:

$$m_{ij}^{(n)} = \text{gMLP}(v'_i || v'_j || e'_{ij}) \quad (11)$$

The fusion feature $m_{ij}^{(n)}$ is transformed nonlinearly to yield the edge update Δe_{ij} . For the node update Δv_i , we first apply the learnable envelope function (Eq. 12), then aggregate, and finally apply a nonlinear transformation.

$$\mu_{ij}^{(n)} = \text{Linear}(e'_{ij}) \quad (12)$$

$$\Delta v_i = \text{MLP}\left(\sum_{k \in \mathcal{N}(i)} (\mu_{ij}^{(n)} \odot m_{ij}^{(n)})\right) \quad (13)$$

$$\Delta e_{ij} = \text{MLP}(m_{ij}^{(n)})$$

In the Readout block, the final atom features $v_i^{(N)}$ are used to directly predict the total energy (E) and magnetic moments (M):

$$\begin{aligned} E &= \sum_i \text{MLP}(v_i^N) + \text{ref}(z_i) \\ M &= \text{MLP}(v_i^N) \end{aligned} \quad (14)$$

Here, $\text{ref}(z_i)$ is obtained by performing a least-squares fit to the dataset energies. The atomic forces F_i and stress (σ) are obtained via automatic differentiation of the energy with respect to the atomic Cartesian coordinates X_i and the lattice strain tensor (ε).

$$F_i = -\frac{\partial E}{\partial X_i}, \quad \sigma = \frac{1}{V} \frac{\partial E}{\partial \varepsilon} \quad (15)$$

B TRAINING STRATEGIES

B.1 LOAD-BALANCE STRATEGY

In most atomic datasets, the sizes of structures typically follow a long-tailed distribution (Deng et al., 2023; Barroso-Luque et al., 2024). Fig 6(a) visualizes the distributions of atom and edge counts in the MPTrj dataset (cutoff radius 6.0). In distributed training, allocating samples with a fixed batch size may cause two issues: (1) a GPU may receive only large samples, leading to out-of-memory (OOM) and limiting the maximum batch size; (2) assuming memory allows, some GPUs may have higher computational loads while others have lower loads, causing idle time and extra synchronization overhead. To address this, we adopt a **load balancing strategy**:

1. Shuffle the entire dataset to ensure randomness, then split it into multiple chunks.
2. Within each chunk, sort samples in descending order of size.
3. Using a greedy algorithm, assign samples sequentially to GPUs, prioritizing the GPU with the largest remaining capacity. If all GPUs are “full” or the chunk has no remaining samples, a batch is generated.

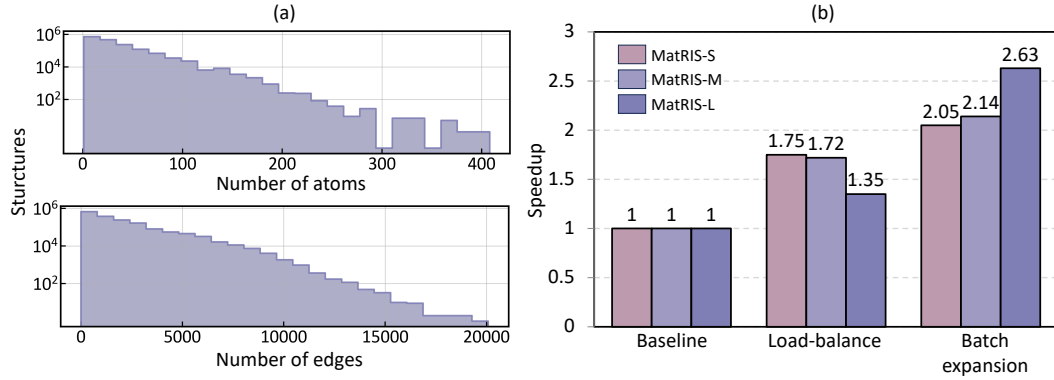


Figure 6: (a) Number of atoms and edges in MPTrj dataset. (b) Training speedup achieved by load balancing.

As shown in Fig. 6(b), with load balancing, the cumulative speedup reaches **1.35–1.75** $\times$, indicating that synchronization overhead due to load imbalance is significant during each training step. Building on this, the global batch size can be increased (denoted as “Batch expansion”), resulting in a cumulative speedup of **2.05–2.63** $\times$.

B.2 LOSS-BALANCE STRATEGY

Besides load imbalance during training, the loss computation can also be imbalanced. This is especially clear for the force and magmom loss (e.g., node-level tasks). For example, the MSE loss (forces only) is defined as:

$$L_f = \frac{1}{3N_B} \sum_{i=1}^{N_B} |f_i^{\text{pred}} - f_i^{\text{DFT}}|^2 \quad (16)$$

where N_B denotes the total number of atoms in this global batch.

This reduction method causes the losses of large samples to dominate during training, effectively making the model prioritize large samples and resulting in a loss imbalance. To address this, we adopt a graph-level loss reduction strategy (referred to as **Graph-level loss**). Specifically, when computing node-level losses such as force or magmom, we first reduce the loss within each graph, and then perform a second reduction across the global batch, thereby mitigating the loss imbalance. Ablation studies in Table. 6(b) demonstrate the effectiveness of this method. The corresponding formula is as follows:

$$L_f = \frac{1}{B} \sum_{b=1}^B \left(\frac{1}{N_b} \sum_{i=1}^{N_b} \|f_i^{\text{pred}} - f_i^{\text{DFT}}\|^2 \right) \quad (17)$$

where B is the number of graphs in this global batch, N_b is the number of atoms in the b -th graph.

B.3 DETAILS OF DENOISING PRETRAINING

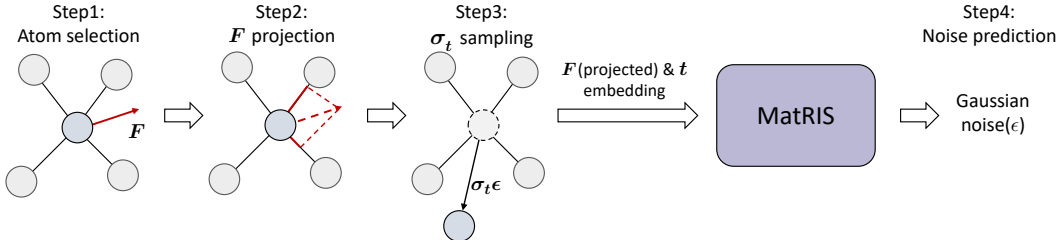


Figure 7: The workflow of denoising pretraining on MatRIS.

In this section, we present the denoising strategy used in our work. Several studies, such as Noisy Node(Godwin et al., 2022), and DeNS(Liao et al., 2024a), have demonstrated that denoising can mitigate the over-smoothing of GNNs and improve generalization. However, these methods have restricted applicability. For example, Noisy Node must be applied to equilibrium structures, while DeNS extends it to non-equilibrium states but is limited to equivariant neural networks. In practice, most mainstream atomic datasets are not fully in equilibrium (Deng et al., 2023; Schmidt et al., 2024; Barroso-Luque et al., 2024), and MLIPs are not necessarily equivariant GNNs. The training of MatRIS faces this scenario. For non-equivariant GNNs on non-equilibrium structures, a key challenge in applying denoising is how to encode forces. Inspired by (Liao et al., 2024a), we project forces onto edges to scalarize the vectors. The denoising workflow is illustrated in Fig. 7.

1. Given a corruption probability, we randomly select atoms to corrupt.
2. For each corrupted atom, we project its force onto the relative position edges by computing inner products, resulting in a scalarized force feature $F(\text{projected})$.
3. We sample a random timestep t and obtain the noise standard deviation σ_t from a linear schedule. We then add noise as $\sigma_t \epsilon$, where ϵ is standard Gaussian noise.
4. The noisy structure, the projected force $F(\text{projected})$, and the timestep t are used as inputs to the MatRIS model, which directly predicts the added Gaussian noise.

After completing the denoising training, we fine-tune the model based on these weights.

C TRAINING DETAILS AND HYPER-PARAMETERS

We summarize the hyperparameters of the models across different datasets and versions in Table 7. For the MPTrj dataset, we experimented with models of different sizes by varying the number of layers (S, M, and L correspond to 4, 6, and 12 layers, respectively). The learning rate follows a cosine annealing schedule, decaying to the minimum value at the last epoch. In distributed training, we implemented a load balancing scheme to reduce synchronization overhead (see Sec. B.1), so the number of samples per GPU may vary; the table reports the average global batch size. Notably, the loss reduction in distributed training is performed using a graph-level scheme to avoid bias toward larger systems (see Sec. B.2). Additionally, for the Matbench-Discovery benchmark, we pre-trained the MatRIS-M and L models with a denoising process to enhance generalization, with details in Sec.B.3.

Table 7: Hyper-parameters for MatRIS variants reported in this paper.

Hyper-parameters	MPTri-S	MPTri-M	MPTri-L	SPICE-M	MatPES-1.4M	Others
Number of MatRIS layers	4	6	10	6	3	6
Dimension of atom features	128	128	128	128	64	128
Dimension of edge features	128	128	128	128	64	128
Dimension of three-body features	128	128	128	128	64	128
Radial basis function	Bessel	Bessel	Bessel	Bessel	Bessel	Bessel
Number of radial basis functions	7	7	7	7	7	7
Pairwise cutoff	6.0	6.0	6.0	6.0	6.0	6.0
Three-body cutoff	4.0	4.5	4.5	4.0	4.0	4.0
Global batch size (avg)	512	512	512	128	64	1/256
Optimizer	AdamW	AdamW	AdamW	AdamW	Adam	AdamW
Weight decay	1e-2	1e-2	1e-2	1e-3	0.0	1e-3
Maximum Learning rate	5e-4	3e-4	3e-4	3e-4	3e-4	3e-4
Minimum Learning rate	5e-6	3e-6	3e-6	3e-6	3e-6	3e-6
Learning rate scheduling	CosLR	CosLR	CosLR	CosLR	CosLR	CosLR
Number of epochs	30	40	100	200	30	100
Loss function	Huber($\delta=0.01$)	Huber($\delta=0.01$)	Huber($\delta=0.01$)	Huber($\delta=0.01$)	MAE	Huber($\delta=0.01$)
Gradient clipping norm	0.05	0.05	0.05	0.05	0.05	0.05
Energy loss prefactor	5	5	5	5	5	5
Force loss prefactor	5	5	5	5	5	5
Stress loss prefactor	0.1	0.1	0.1	-	0.1	0.1
Magmom loss prefactor	0.1	0.1	0.1	-	0.1	-
Denoising Settings						
Corruption probability	-	100%	100%	-	-	-
Maximum number of steps	-	1000	1000	-	-	-
Maximum sigma	-	1.0	1.0	-	-	-
Minimum sigma	-	0.0	0.0	-	-	-
Noise Schedule	-	Linear	Linear	-	-	-
Number of epochs	-	20	20	-	-	-

D OTHER DATASET

D.1 ZEOLITE DATASET

The Zeolite Dataset maintained by the International Zeolite Association (IZA) is a comprehensive and authoritative repository of zeolite framework structures. It contains all framework types approved by the IZA Structure Commission, accompanied by crystallographic data, topology drawings, simulated X-ray powder diffraction patterns, framework chemical composition, and references to original literature.

D.2 MATBENCH-DISCOVERY RESULTS FOR FULL AND 10K MOST STABLE SPLITS

Matbench-Discovery is designed to simulate a real high-throughput materials discovery task. Two evaluation splits(full and 10k most stable splits) are also informative.

- **Full (WBM) test set:** the entire held-out set of candidate inorganic structures above the convex hull.
- **10 k most stable split:** for each model, consider the 10,000 structures it predicts to be most stable, and evaluate classification accuracy among that subset. (This measures how well the model ranks top candidates.)

D.3 PHONON PREDICTION

Table 8: MatRIS performance on Zeolite dataset. Energy(**E**) MAE is in **meV**, force(**F**) MAE is in **meV/Å**.

	Deep Pot		AlphaNet		DPA3-L24		MatRIS-M	
Type	E	F	E	F	E	F	E	F
ABWopt	90	90	<u>12</u>	<u>19</u>	13.7	19.1	3.4	6.8
BCTopt	110	50	<u>6.8</u>	<u>12</u>	<u>6.8</u>	<u>12.0</u>	3.0	4.9
BPHopt	210	60	<u>29</u>	16	28.6	<u>16.5</u>	39.2	8.3
CANopt	150	90	18	<u>15</u>	<u>17.3</u>	15.1	10.2	6.5
EDIopt	40	50	<u>8.7</u>	<u>13.5</u>	10.8	15.1	3.6	6.2
FERopt	290	130	43	28	<u>41.6</u>	<u>25.4</u>	30.6	11.9
GISopt	60	50	<u>11</u>	<u>11</u>	12.1	11.8	6.2	5.8
JBWopt	150	70	10	13	<u>9.7</u>	<u>12.4</u>	4.0	5.9
LOSopt	110	70	<u>21</u>	<u>12</u>	<u>21.0</u>	13.1	17.9	6.5
LTAopt	150	64	19	<u>12</u>	21.3	13.5	<u>20.5</u>	7.0
LTJopt	70	110	15	<u>11</u>	13.4	11.2	<u>14.4</u>	6.7
NATopt	210	110	<u>16</u>	<u>15</u>	17.2	15.6	8.0	7.4
PARopt	90	70	<u>18</u>	<u>24</u>	21.4	24.2	9.5	6.8
PHIopt	60	120	20	<u>14</u>	<u>19.6</u>	14.4	15.9	7.5
SODOpt	200	110	<u>9.5</u>	<u>12</u>	11.0	13.4	4.4	5.7
THOopt	160	60	<u>17</u>	<u>15</u>	20.2	16.9	7.5	8.0

Table 9: MatRIS performance on the compliant Matbench-Discovery benchmark with results on **full test set**. ‘↑’/‘↓’ stands for higher/lower is better. All models accessed before September 24, 2025.

Model	Param.	F1↑	DAF↑	Precision↑	Recall↑	Accuracy↑	MAE↓	R2↑
CHGNet	0.41M	0.612	3.038	0.521	0.740	0.839	0.061	0.690
MACE-MP-0	4.69M	0.668	3.400	0.583	0.781	0.867	0.055	0.698
GRACE-2L	15.3M	0.687	3.714	0.637	0.746	0.884	0.050	0.740
Allegro-MP-L	18.7M	0.742	4.998	0.686	0.809	0.904	0.042	0.775
Nequix MP	0.71M	0.744	3.950	0.678	0.825	0.903	0.042	0.780
SevenNet-l3i5	1.17M	0.751	4.112	0.706	0.803	0.909	0.042	0.773
NequIP-MP-L	9.6M	0.752	4.136	0.710	0.800	0.909	0.041	0.787
ORB v2 MPtrj	25.2M	0.755	4.188	0.719	0.795	0.911	0.043	0.756
HIENet	7.51M	0.766	4.361	0.748	0.84	0.918	0.039	0.791
Eqnorm MPtrj	1.31M	0.779	4.296	0.737	0.825	0.920	0.038	0.796
DPA-3.1-MPtrj	4.81M	0.789	4.419	0.758	0.822	0.924	0.036	0.809
eqV2 S DeNS	31.2M	0.798	4.362	0.748	0.855	0.927	0.035	0.785
eSEN-30M-MP	30.1M	<u>0.819</u>	4.616	0.792	<u>0.849</u>	0.936	<u>0.032</u>	<u>0.818</u>
MatRIS-S	4.3M	0.798	4.522	0.776	0.821	0.929	0.035	0.800
MatRIS-M	6.3M	<u>0.819</u>	<u>4.713</u>	<u>0.809</u>	0.830	<u>0.938</u>	<u>0.032</u>	0.817
MatRIS-L	10.4M	0.829	4.765	0.818	0.841	0.941	0.030	0.825

Table 10: MatRIS performance on the compliant Matbench-Discovery benchmark with results on **10k most stable**. ‘↑’/‘↓’ stands for higher/lower is better. All models accessed before September 24, 2025.

Model	Param.	F1↑	DAF↑	Precision↑	Recall↑	Accuracy↑	MAE↓	R2↑
MACE-MP-0	4.69M	0.888	5.221	0.798	1.000	0.798	0.087	0.508
GRACE-2L	15.3M	0.914	5.503	0.841	1.000	0.851	0.065	0.641
Allegro-MP-L	18.7M	0.917	5.536	0.846	1.000	0.846	0.064	0.706
CHGNet	0.41M	0.920	5.567	0.851	1.000	0.851	0.063	0.816
SevenNet-13i5	1.17M	0.952	5.945	0.909	1.000	0.909	0.050	0.745
Nequix MP	0.71M	0.953	5.949	0.910	1.000	0.910	0.051	0.734
NequIP-MP-L	9.6M	0.957	6.000	0.917	1.000	0.917	0.044	0.812
Eqnorm MPtrj	1.31M	0.970	6.156	0.941	1.000	0.941	0.039	0.775
ORB v2 MPtrj	25.2M	0.971	6.173	0.944	1.000	0.944	0.037	0.801
HIENet	7.51M	0.973	6.194	0.947	1.000	0.947	0.033	0.894
eSEN-30M-MP	30.1M	0.978	6.261	0.957	1.000	0.957	0.035	0.755
DPA-3.1-MPtrj	4.81M	0.980	6.288	0.961	1.000	0.961	0.032	0.838
eqV2 S DeNS	31.2M	0.983	6.326	0.967	1.000	0.967	0.031	0.823
MatRIS-S	4.3M	0.987	6.367	0.973	1.000	0.973	0.025	0.937
MatRIS-M	6.3M	<u>0.989</u>	<u>6.393</u>	<u>0.977</u>	1.000	<u>0.977</u>	<u>0.021</u>	<u>0.945</u>
MatRIS-L	10.4M	0.990	6.418	0.981	1.000	0.981	0.020	0.948

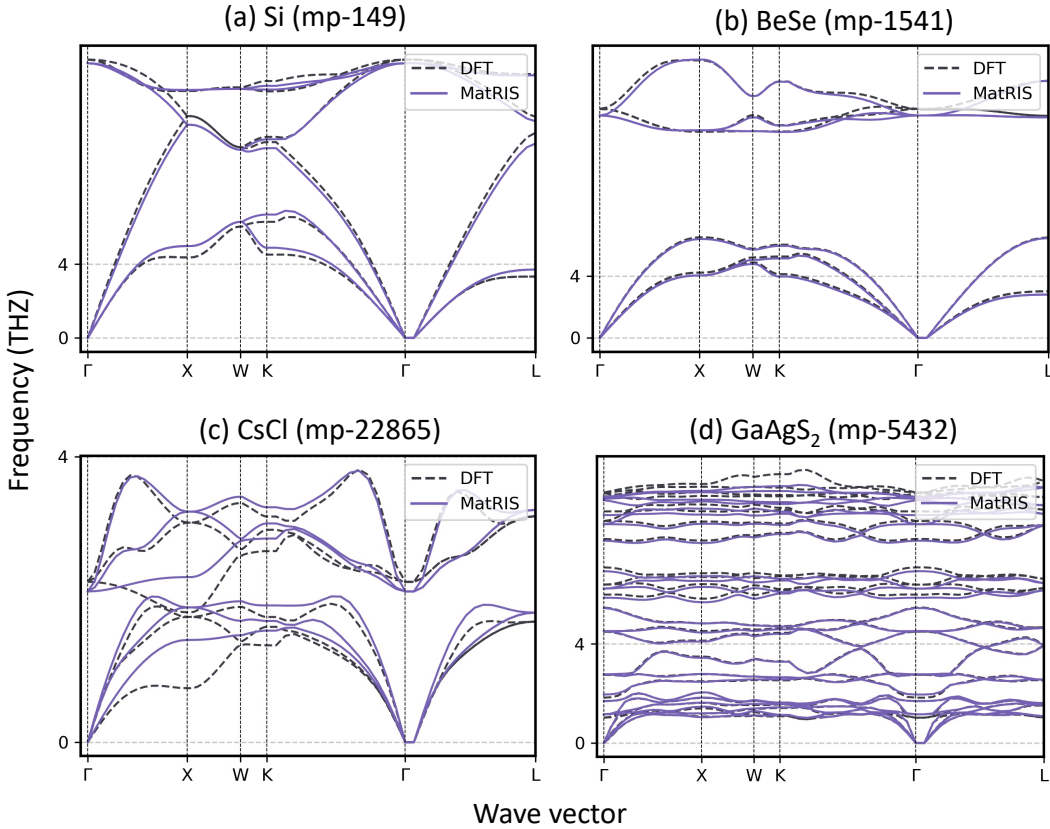


Figure 8: Comparison of phonon dispersion and DOS for (a) Si, (b) BeSe, (c) CsCl, and (d) GaAgS₂, as predicted by MatRIS, and DFT calculations from PhononDB.